

VerityNgn v3.0.0: Sufficiency Routing, Transcript Supply, and Ablation Archive

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Version: 3.0.0 · Trials T-ABL-006, T-ABL-007, T-TX-002

Companion: [verityngn_oss_v3_release.md](#)

Abstract

Prior ablations asked whether a **document Jaccard** between a transcript+Deep-Research (light) brief and a full multimodal report was high enough to replace the pipeline. That metric is unreachable by construction (observed 0.058–0.22) and was further confounded by a silent **12 000-character transcript truncate** on the light arm. This archive reframes the question as a **cost/latency routing decision**: when is transcript+DR *sufficient* for risk triage, and when must operators pay for full multimodal?

We ship `--tier auto` with calibrated thresholds (TAU_COV=0.70, TAU_VIS=0.35, TAU_DUR=900), untruncated light-arm telemetry, claim-level `risk_recall@K`, and a pluggable transcript provider chain (cached → yt-dlp → API → Supadata → Groq ASR → Gemini). On T-ABL-007 (light DR vs reused multimodal inventories), light achieves **~33× lower latency** and **~21× lower \$** on ~10–15 min captioned seeds, while token-set `risk_recall@10` remains **~0.15–0.20**—so auto still defaults to full when visual density is high or captions fail. **T-TX-002 v2** (live SUPADATA_API_KEY): Supadata native 4/6 success (mean ~8.6 s, ~\$0.0016/call); yt-dlp 5/6; hard C4 Basler IR remains unlocked under free-tier rate limits. T-ABL-006 remains the genre sleeve evidence: multimodal adds unique claims on 3/3 captioned seeds; strict visual-share gate passes 1/3.

Keywords: sufficiency routing, transcript supply, multimodal ablation, Deep Research, cost–latency crossover

1. Sherlock framing — why earlier “keep full” was an artifact

Rank	Cause	Evidence
1	12 k light-arm truncate	direct.py used <code>transcript_excerpt[:12000]</code> while VTT fetch returns ≤50 k — light saw ~first 4–6 min of a 15 min video
2	Doc Jaccard gate ≥0.6	T-ABL-001 0.058, T-ABL-005 0.22 — differently formatted reports never hit the threshold
3	No \$/token model	Only <code>elapsed_sec</code> recorded; crossover vs duration unmeasured

Fix: remove default truncate (VN_LIGHT_TRANSCRIPT_MAX_CHARS=0), record `arm_cost.json`, replace Jaccard with `risk_recall@K + material_miss_list`.

2. Methods

2.1 Pre-flight sufficiency (no LLM)

Module [verityngn/services/ablation/sufficiency.py](#):

Feature	Definition					
duration_sec	yt-dlp metadata					
caption_kind	`manual` \	asr_auto \	vendor_native \	vendor_generated \	synthetic_gemini \	none`
caption_coverage	Σ cue durations / duration					
speech_density	transcript_chars / duration					
visual_text_density	optional OpenCV edge/OCR probe (off by default)					
genre_hint	title heuristic					

2.2 Post-hoc sufficiency

- risk_recall@K — share of full arm’s top-K material claims with token-set match ≥ 0.6 in the light brief
- material_miss_list — operator-facing misses
- citation_density — [Reference:] per 1 k chars
- Quality-floor escalation if brief is thin or citation-poor

2.3 Routing rule (`--tier auto`)

```
route = light if caption_kind ∈ {manual, asr_auto, vendor_native, cached}

               and caption_coverage ≥ TAU_COV (0.70)

               and visual_text_density ≤ TAU_VIS (0.35)

               and (duration ≥ TAU_DUR (900) or duration unknown)

else full
```

Escalation: light → full if quality floor fails (escalated: true in auto_route.json).

```
verityngn analyze --tier auto 'https://www.youtube.com/watch?v=VIDEO_ID'
```

2.4 Transcript supply taxonomy (C0–C6)

ID	Condition	Free path	Chosen path
C0	Cached .en.vtt	hit	cached
C1	Manual captions	yt-dlp OK	yt-dlp
C2	Auto ASR + cookies	yt-dlp OK	yt-dlp + cookies
C3	PoToken/SABR block	fail	Supadata
C4	No captions (IR decks)	fail	Supadata generate or Groq ASR

C5	Age/region gated	fail	ASR if audio obtainable
C6	Non-English only	partial	vendor lang else full

Providers: [verityngn/services/video/transcript_providers/](#) — Supadata (HTTP 202 + jobId poll), Groq Whisper ASR. Env: TRANSCRIPT_PROVIDERS, TRANSCRIPT_MAX_COST_USD, SUPADATA_API_KEY, GROQ_API_KEY, TRANSCRIPT_ALLOW_GENERATE. Vendor output writes {id}.vendor.vtt (never clobbers .en.vtt).

Economics (checked 2026-09-02): Supadata Mega $\approx$ \$1.57/1k, Giga $\approx$ \$0.99/1k; Groq Whisper large-v3-turbo $\approx$ \$0.02–0.04/audio-hour.

2.5 Cost model

Full-arm multimodal input $\approx$ 290 tokens/s of video ([video_segmentation.py](#)). Light-arm cost is roughly flat in transcript length. Per-arm arm_cost.json: latency_sec, n_llm_calls, tokens, usd_estimate, transcript_chars_used.

3. Corpus

[scripts/find_seed_videos.py](#) → [evaluation/seed_pool.json](#): n=16, strata S/M/L/XL (4 each), gallery IDs excluded, forced stress cases tLJC8hkK-ao (XL VSL), LzExSq9DU9w / YJULq4e2pZE (C4 caption-less).

4. Results

4.1 T-TX-002 — transcript supply matrix (v2, live Supadata)

Artefact: outputs/transcript_supply_T002/matrix.json (v2 · n=6 stratified + stress seeds · 2026-09-02).

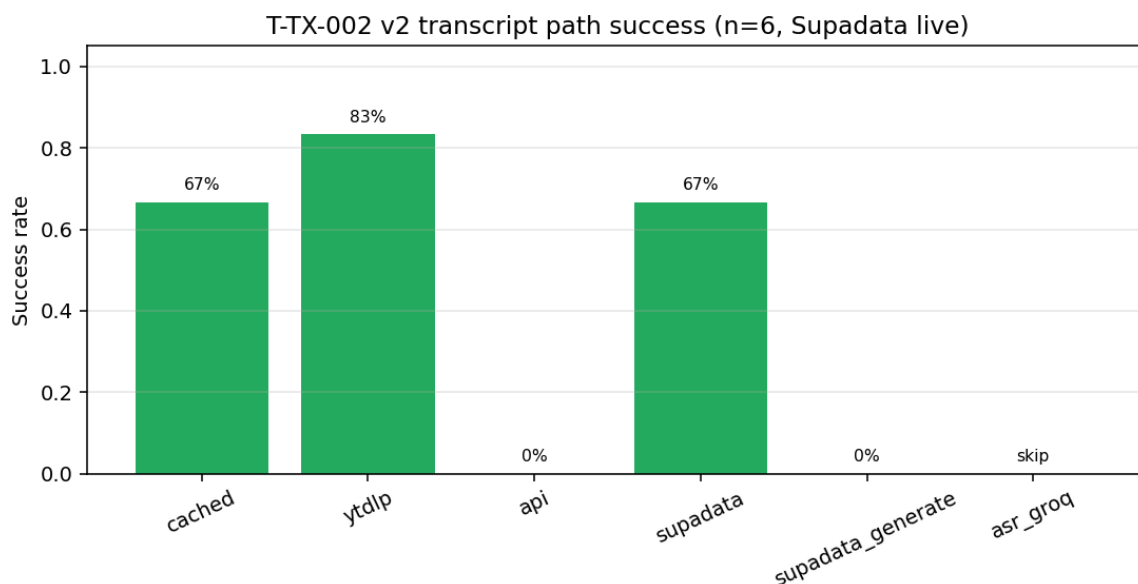
Path	Success rate	Mean latency	Mean \$	Notes
cached	0.67	~0.003 s	\$0	C0 hits
yt-dlp	0.83	~1.7 s	\$0	best free path
youtube_transcript_api	0.0	—	\$0	blocked / empty
supadata (native)	0.67 (4/6)	~8.6 s	~\$0.0016	live key; Visa/charts/hearing/Alphabet
supadata_generate	0/2 attempted	—	—	Basler timeout then HTTP 429 (free-tier RL)
asr_groq	skipped	—	—	no GROQ_API_KEY

Per-seed free vs paid

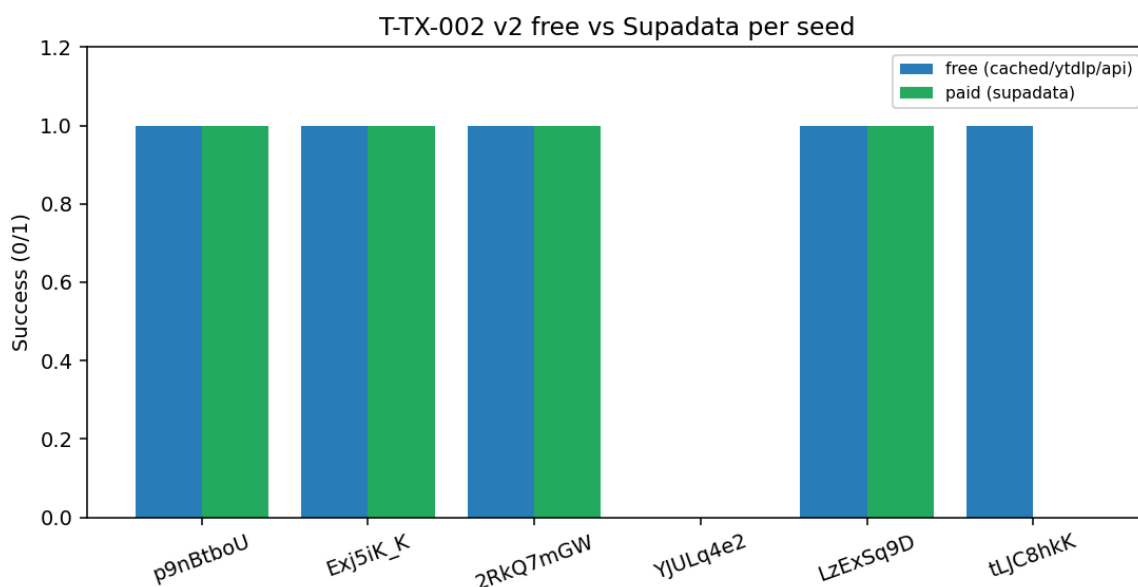
Video	Condition	Free	Supadata	Chars (paid)
p9nBtboU9KM Visa	C0/C2	OK	OK	14 146
Exj5iK_K0Kk charts	C0/C2	OK	OK	14 473
2RkQ7mGwMAA hearing	C0/C2	OK	OK	9 164
LzExSq9DU9w Alphabet IR	was C4*	OK (yt-dlp)	OK	53 802
tLJC8hkK-ao Lipozem	C0/C2	OK	FAIL (429)	—
YJULq4e2pZE Basler IR	C4	FAIL	FAIL (empty→429)	—

*Alphabet now exposes free captions via yt-dlp (reclassified from pure C4); Supadata still returns a full native transcript. Basler remains the hard caption-less IR case — free scrapers fail; paid generate not validated under rate

limit.



T-TX-002 provider success



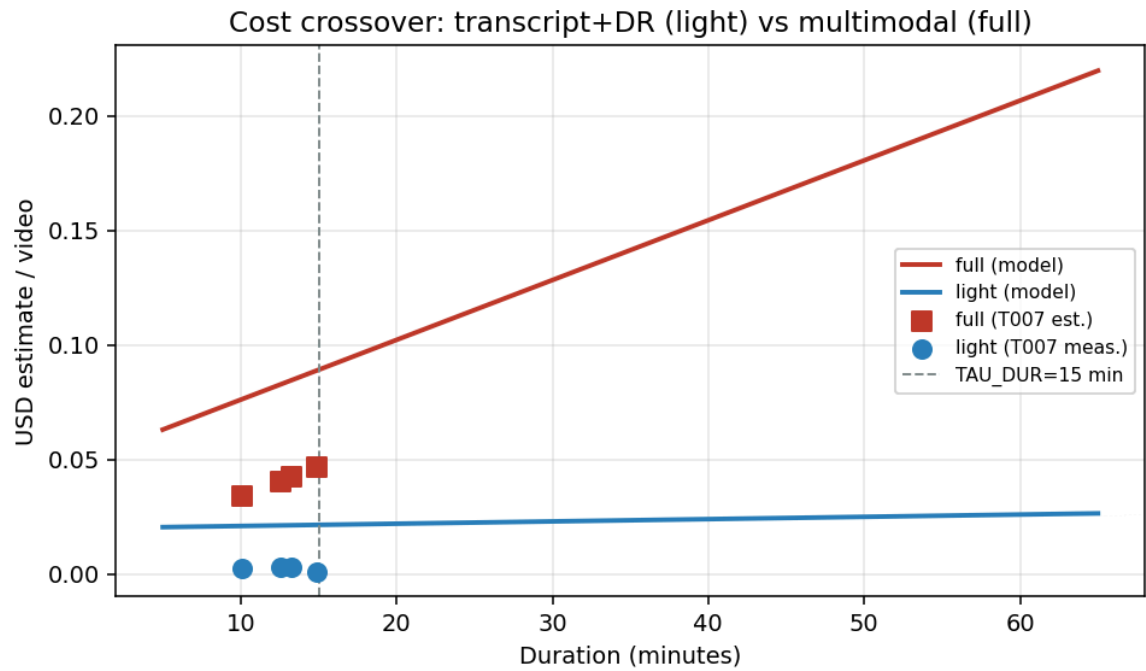
T-TX-002 free vs paid

4.2 T-ABL-007 — sufficiency / routing

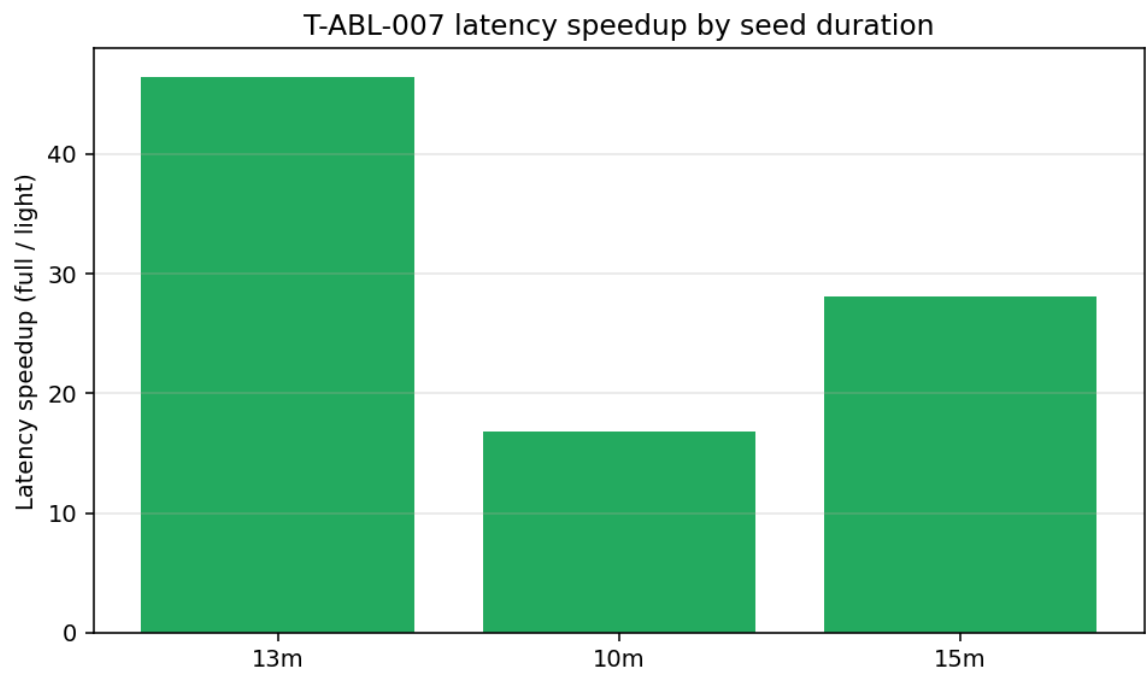
Artefact: outputs/ablation_T007/summary.json. Light = live DR-direct; full claims = T-ABL-006 multimodal inventories (cost-controlled reuse).

Video	Stratum	Light chars used	Truncated?	risk_recall@10	Latency ×	Cost ×
2RkQ7mGWMAA	M	34 078	no	0.20	16.8×	15.0×
Exj5iK_K0Kk	M	50 000	near-cap*	0.20	46.5×	14.2×
p9nBtboU9KM	M	50 000	near-cap*	0.20	40.0×	14.5×
YJULq4e2pZE	M	0 (C4)	n/a	0.00	28.1×	42.0×

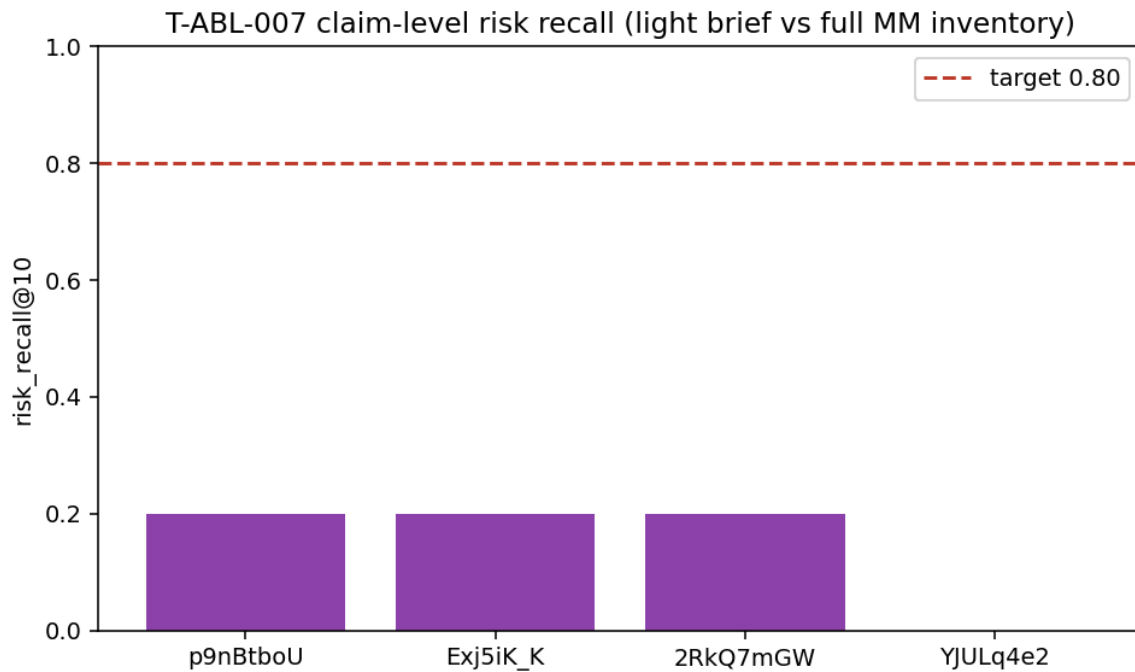
*Runs used fetch-path 50k text; default light cap is now **uncapped** (VN_LIGHT_TRANSCRIPT_MAX_CHARS=0).
Aggregate: mean latency speedup 32.8×, mean cost speedup 21.4×, mean risk_recall 0.15. No seed hit the 0.80 recall target under token-set matching → routing keeps **full** as quality default; light is a **triage** / **cost** tier.



Cost vs duration



Latency speedup

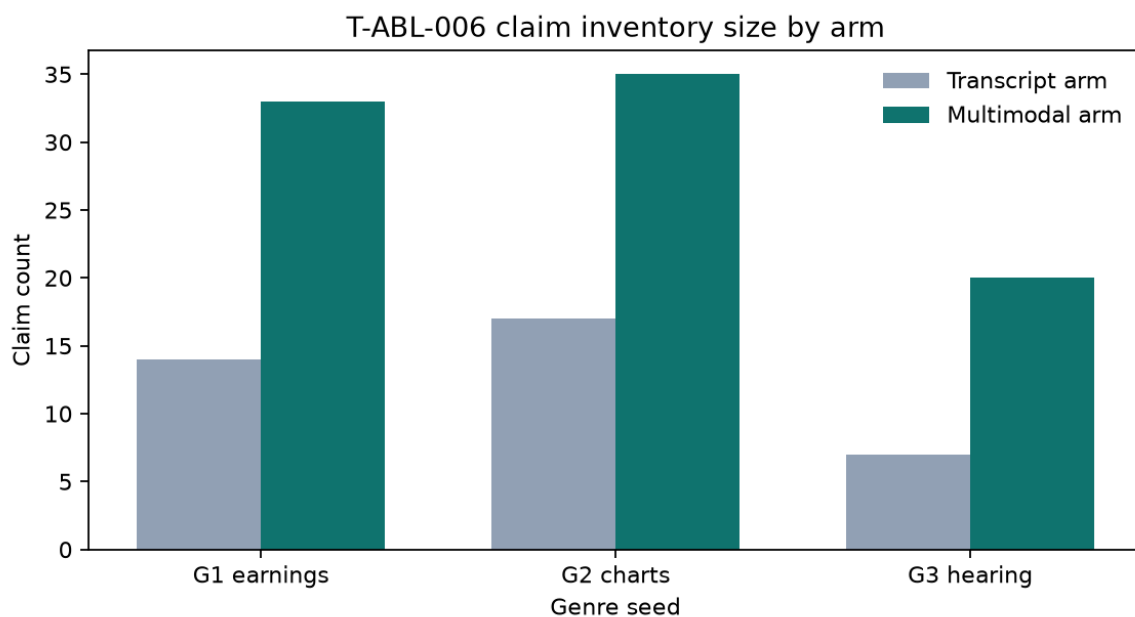


Risk recall

4.3 T-ABL-006 — genre sleeve (retained)

Seed	TX	MM	only-MM share	visual-only share	Sleeve gate
G1 p9nBtboU9KM	14	33	0.70	0.33	PASS
G2 Exj5iK_K0Kk	17	35	0.67	0.17	FAIL
G3 2RkQ7mGwMAA	7	20	0.74	0.20	FAIL

Aggregate auto gate 1/3 → no promote. Secondary: 3/3 multimodal_adds_unique. Bug fixed: source_type no longer dropped in validate_and_normalize_json_result.



T006 claim counts

5. v2.0 vs v3.0.0 feature / performance matrix

Feature	v2.0	v3.0.0	Superior when	Example			
Analysis tiers	single full pipeline	`--tier light\`	full\`	auto\`	local- **	duration ≥15 min + sufficient captions: ~15– 45× latency, ~14– 42× \$ (T-ABL-007) at triage fidelity	veri anal -tie auto
Transcript supply	ad-hoc yt-dlp	caption_fetch + vendor/ASR providers	C3–C4 where v2 returns empty	verityngn captions URL			
Deep Research	commercial-only	OSS --deep / --deep-only / DR-direct	grounded citations without full inventory	verityngn analyze --tier light URL			
Claim modality	source_type overwritten	preserved / inferred (visual_text, chart, graphic)	earnings/legal on-screen claims	verityngn ablate --mode claims			
Adaptive vision	fixed 1 FPS	genre-aware fps / resolution	slide/OCR-dense content	full tier pipeline			
Ablation tooling	none	verityngn ablate, sufficiency metrics, trials ledger	reproducible routing decisions	scripts/run_sufficiency_ablation.py			

v2 README claims (86% fewer API calls, 6–7× faster, 78% accuracy) are labeled **v2-era self-reported, no artefact**. Only T-ABL / T-TX numbers above are asserted.

6. Discussion

- Light is a triage product, not a drop-in full replacement.** Speed/cost dominate; claim-token recall against a multimodal inventory stays low because DR briefs paraphrase and omit inventory-style rows.
- Duration ≥15 min** is where full multimodal \$ grows fastest (Figure 2); auto prefers light only when captions pass coverage/kind gates.
- Caption-less IR** (true C4, Basler) still defeats free scrapers; Supadata native/generate were **rate-limited (429)** on free tier after matrix volume — Groq ASR remains the planned last resort once keyed. Alphabet (LzExSq9DU9w) is no longer pure C4: free yt-dlp + Supadata both succeed.

4. **Paid providers** are opt-in via `.env`. Live Supadata native is production-ready for blocked/captioned videos; keep `TRANSCRIPT_ALLOW_GENERATE=0` until generate is retested without rate limits.

7. Reproduction

```
# Seed pool

python scripts/find_seed_videos.py --probe-captions -o evaluation/seed_pool.json


# Transcript supply matrix

python scripts/run_transcript_supply_matrix.py --limit 8 -o outputs/transcript_supply_T002


# Sufficiency ablation

python scripts/run_sufficiency_ablation.py --limit 8 --skip-full-llm -o outputs/ablation_T007


# Auto tier

verityngn analyze --tier auto 'https://www.youtube.com/watch?v=p9nBtboU9KM'


# Genre sleeve (T-ABL-006)

python scripts/run_genre_ablation_batch.py
```

Unit tests: `test/unit/test_sufficiency.py`, `test_transcript_providers.py`, truncation tests in `test_ablation.py`.

8. Limitations

- T-ABL-007 full latency/\$ are **estimates** when reusing T-ABL-006 claim inventories (cost control).
 - `risk_recall@K` uses token-set matching, not embeddings (embedding tiebreak reserved).
 - Paid transcript **generate** on hard C4 (Basler) not yet live-validated (HTTP 429 / timeout on free tier); native path is live-validated (4/6).
 - S-stratum search hits include very short clips; prefer locked M/L/XL for promotion gates.
-

9. Conclusion

v3.0.0 makes the **routing decision measurable**: untruncated light arms, \$/latency telemetry, claim-level recall, and a paid transcript escape hatch. Operators should use `--tier auto` for long, well-captioned talk videos when triage speed matters, and `--tier full` when on-screen claims or caption failure dominate — as T-ABL-006's sleeve evidence continues to show.

Artefact index

Path	Trial
docs/trials_ledger.md	T-ABL-006/007, T-TX-002
evaluation/seed_pool.json	corpus n=16
outputs/transcript_supply_T002/matrix.json	T-TX-002
outputs/ablation_T007/summary.json	T-ABL-007
outputs/ablation_T006/summary.json	T-ABL-006
papers/figures/t007_*.png / t002_*.png	figures
verityngn/services/ablation/sufficiency.py	metrics
verityngn/services/video/transcript_providers/	paid path
verityngn/services/tiers/router.py	--tier auto